

WASP-10b

R-1875

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_150921 · created 2026-08-03T11:17:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457286.6673 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.055
Transit depth (Rp/Rs)^2	1.45 +/- 1.34 [%]
Orbital Inclination (inc)	84.32 +/- 1.9
Airmass coefficient 1 (a1)	0.315 +/- 0.074
Airmass coefficient 2 (a2)	0.91 +/- 0.12
Scatter in the residuals of the lightcurve fit is	20.88 %
Best Comparison Star	#1 - [389, 361]
Optimal Method	PSF photometry
Transit Duration (day)	0.03 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.121 ± 0.055 vs 0.1592 (deviation 0.48σ)
Duration fit vs published	0.03 d vs 0.0946 d (deviation 1.36σ)
Mid-time O-C	-6.1 min
Depth SNR	1.5
Residual scatter	20.88 %

Light curve

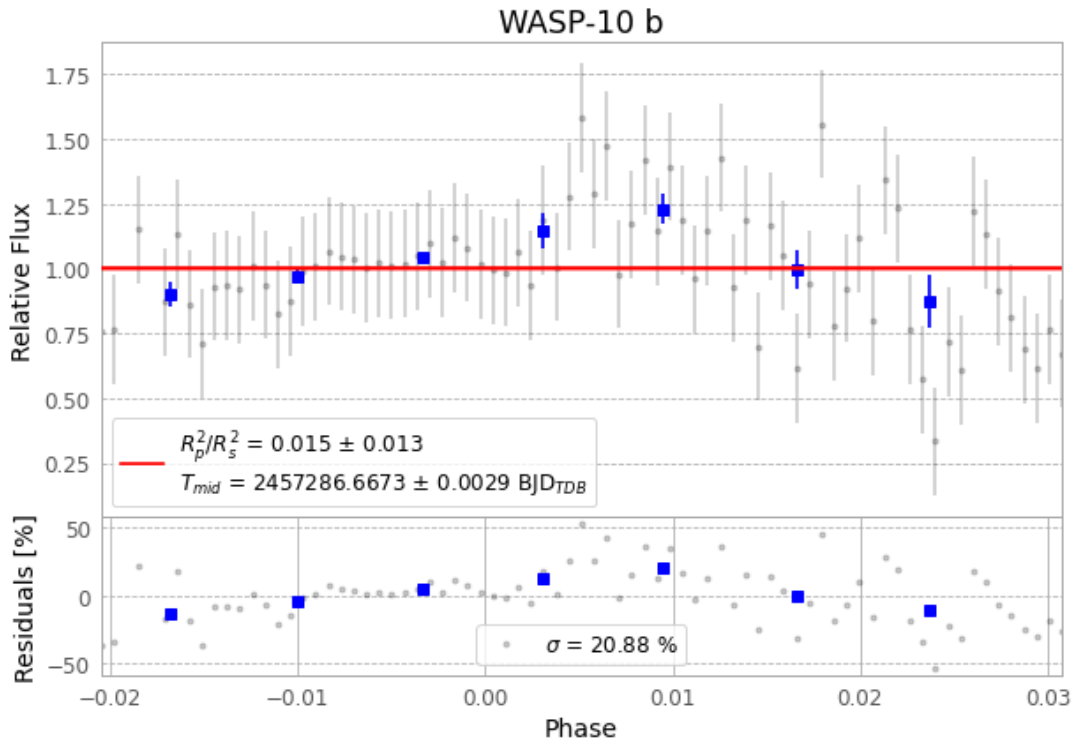


Figure 1 — FinalLightCurve_WASP-10 b_2015-09-20.png (SHA-256 aae9768c925462d1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [505, 133], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17e76d27581b4f72...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-10150921022112.FITS → WASP-10150921060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-150921.log (d0ef7982cce1...), FinalLightCurve_WASP-10 b_2015-09-20.png (aae9768c9254...), FinalLightCurve_WASP-10 b_2015-09-20.csv (8a8fa3a2753a...)

Compute resources

Wall time 375.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:17Z.